INCH-POUND

A-A-51310B
15 May 1992
SUPERSEDING
A-A-51310A
20 August 1990

COMMERCIAL ITEM DESCRIPTION

IMPRESSION MATERIAL, DENTAL (Modeling Compound)

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers dental impression material (modeling compound) for use in taking impressions in the oral cavity.

Salient characteristics:

Classification: The impression material shall be furnished in the following forms and colors:

Form 1 - Cakes
Color A - Red (brick)

Form 2 - Sticks
Color A - Red (brick)
Color B - Gray
Color C - Green

The impression material shall be a reversible thermoplastic impression compound used to take preliminary impressions of oral structures. The compound shall be a composition of resins and fillers with necessary additives such as plasticizers and coloring agents to ensure proper working temperature range. The impression material shall be Type I, impression compound, in accordance with ANSI/ADA Specification No. 3 for Dental Impression Compound with the additions and exceptions specified herein.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Physical characteristics:

The impression material shall be in the form of a cake or stick. The impression material in stick form shall be suitable for copper band impressions or tracing or adding to the periphery of an impression, and shall be green, gray or red (brick), as specified. The sticks shall be 4 1/2 ± 1/4 inches long and 5/16 ± 1/8 inch in diameter. Cakes shall be suitable for full or partial impressions and shall be red (brick) in color. The form of the cake shall be approximately trapezoidal in shape with slightly concave sides and rounded corners. The width of the cake shall be 3 1/4 ± 1/4 inches, the length shall be 2 ± 1/4 inches and the thickness shall be 3/16 ± 1/16 inch.

Each unit package shall contain 1/2 pound of cake form or 1/4 pound of stick form impression material.

Working characteristics: The impression material shall soften to a workable consistency within the temperature range specified in Table I for the applicable color, when tempered in a suitable water bath.

TABLE I. Working Range

COLOR	WORKING RANGE
Green	122° to 124° F (50.0° to 51.0°C)
Gray	128° to 130° F (53.0° to 54.0°C)
Red	132° to 133° F (55.5° to 56.0°C)

The working time of the impression compound shall be from 60 to 80 seconds when tested as specified herein.

ADA acceptability. The resin shall be listed as "acceptable" by the Council on Dental Materials, Instruments, and Equipment of the ADA.

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Toxicity. Prior to delivery, the manufacturer shall furnish a statement that the impression compound, when used according to the directions supplied, has no toxic effect on a person. The statement shall be sent directly to the Technical Services Branch (Code MSTS-1), Directorate of Medical Materiel, Defense Personnel Support Center, 2800 S. 20th Street, Philadelphia, PA 19101.

Certification. Suppliers shall furnish at time of award a dated certificate and data showing that the impression compound they propose to furnish is the same in quality and brand as that appearing on the list of certified materials for American Dental Association Specification No. 3 for Dental Impression Compound, Type I.

Compliance with this certification requirement does not absolve the contractor from complete compliance with the requirements of this specification.

Instructions. Instructions for use of the impression material, including method of softening, working temperature, and curve or data showing the shrinkage of the compound from 40°C to 20°C shall be supplied with each package. In addition, the contractor shall mail two copies of the instructions to the contracting officer for distribution.

Marking. Each unit container shall be marked with a lot or batch number and date of manufacture.

Workmanship. The impression material shall be free from defects which may detract from its appearance or may impair its serviceability.

Unit. Package (PG). One package containing $^1/_2$ pound of cake form or $^1/_4$ pound of stick form impression material, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after the delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one quarter pound of stick form or one half pound of cake form impression material. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, except for dimensions, which shall be conducted at inspection level S-1, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1. For tests, the acceptance number shall be zero (0) for all sample sizes.

Tests. Tests shall be conducted to determine compliance with specification requirements.

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Working time. This test shall be conducted in a 23°C ± 2°C room. A water bath shall be adjusted to and maintained at the working range temperature of the compound (see Table I). A 30 ± 2 gram sample of the impression compound shall be placed in the water bath and warmed for a period of 15 minutes. The warm sample shall then be removed from the water bath and worked (kneaded) in wet hands for 15 seconds. The sample shall then be returned to the water bath for an additional five minutes of conditioning. The sample shall then be removed from the water bath again and worked (kneaded) in wet hands until it loses plasticity, or proper working consistency, and can no longer be molded by finger pressure. The total elapsed time from final removal from the water bath to the loss of plasticity is the working time.

Weight. The weight of the impression material shall be determined using a suitable balance.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Numbers	Item Identification
6520-00-536-2710	Impression Material, Dental Cakes, Red, 1/2 lb (226.8 Grams)
6520-00-765-0251	Impression Material, Dental, Sticks, Gray, 1/4 lb (113.4 Grams)
6520-01-020-1427	Impression Material, Dental Sticks, Green, 1/4 lb (113.4 Grams)
6520-01-020-1428	Impression Material, Dental, Sticks, Red, 1/4 lb (113.4 Grams)

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MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6520-2521
Location: PL Disk #2/IMPRESS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I. RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-51310B	92-05-15

3. DOCUMENT TITLE
IMPRESSION MATERIAL, DENTAL (Modeling Compound)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
Defense Personnel Support Center	(1) Commercial: (215)737-2870 (2) AUTOVON: 444-2870
c. ADDRESS (Include Zip Code) ATTN: DPSC-MSE 2800 S. 20th Street Philadelphia, PA 19101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 — Previous editions are obsolete — 198/290

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